

1.5: Further Vectors

1. Scalar (Dot) Product — Recap

$$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3 = |\mathbf{a}||\mathbf{b}| \cos \theta$$

- Scalar result. Commutative.
- $\mathbf{a} \cdot \mathbf{b} = 0$ iff $\mathbf{a}$ and $\mathbf{b}$ are perpendicular (or one is zero).
- Used to find angle between two vectors.

2. Vector (Cross) Product

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = (a_2b_3 - a_3b_2)\mathbf{i} - (a_1b_3 - a_3b_1)\mathbf{j} + (a_1b_2 - a_2b_1)\mathbf{k}$$

Key Properties

- Vector result, perpendicular to both $\mathbf{a}$ and $\mathbf{b}$.
- Magnitude $|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}||\mathbf{b}| \sin \theta$ — area of the parallelogram spanned by $\mathbf{a}$ and $\mathbf{b}$.
- Direction given by the **right-hand rule**: curl fingers from $\mathbf{a}$ to $\mathbf{b}$, thumb gives direction.
- **Anti-commutative**: $\mathbf{b} \times \mathbf{a} = -(\mathbf{a} \times \mathbf{b})$.
- $\mathbf{a} \times \mathbf{b} = \mathbf{0}$ iff $\mathbf{a}$ and $\mathbf{b}$ are parallel (or one is zero).
- Distributive over addition: $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$.

Area of triangle with vertices $A(1, 0, 2)$, $B(3, 1, 1)$, $C(0, 2, 4)$.

$$\overrightarrow{AB} = (2, 1, -1), \overrightarrow{AC} = (-1, 2, 2).$$

$$\overrightarrow{AB} \times \overrightarrow{AC} = (1 \cdot 2 - (-1) \cdot 2, -[2 \cdot 2 - (-1) \cdot (-1)], 2 \cdot 2 - 1 \cdot (-1)) = (4, -3, 5).$$

$$\text{Magnitude} = \sqrt{16 + 9 + 25} = \sqrt{50} = 5\sqrt{2}.$$

$$\text{Area} = \frac{1}{2} |\overrightarrow{AB} \times \overrightarrow{AC}| = \frac{5\sqrt{2}}{2}.$$

3. Scalar Triple Product

$$[\mathbf{a}, \mathbf{b}, \mathbf{c}] := \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \det \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix}$$

- Scalar result.
- $|\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$ = volume of the parallelepiped spanned by the three vectors.
- $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \mathbf{b} \cdot (\mathbf{c} \times \mathbf{a}) = \mathbf{c} \cdot (\mathbf{a} \times \mathbf{b})$ (cyclic).
- Zero iff the three vectors are **coplanar**.

4. Equation of a Line in 3D

Vector Form

$$\mathbf{r} = \mathbf{a} + \lambda \mathbf{d}, \quad \lambda \in \mathbb{R}$$

where $\mathbf{a}$ is the position vector of any point on the line and $\mathbf{d}$ is a direction vector.

Cartesian Form

Eliminate the parameter from $x = a_1 + \lambda d_1$, etc.:

$$\frac{x - a_1}{d_1} = \frac{y - a_2}{d_2} = \frac{z - a_3}{d_3} \quad (\text{when all } d_i \neq 0)$$

5. Equation of a Plane

Vector (Normal) Form

$$(\mathbf{r} - \mathbf{a}) \cdot \mathbf{n} = 0, \quad \text{equivalently} \quad \mathbf{r} \cdot \mathbf{n} = d$$

where $\mathbf{n}$ is normal to the plane and $\mathbf{a}$ lies on it.

Cartesian Form

$$n_1x + n_2y + n_3z = d$$

Parametric Form

$$\mathbf{r} = \mathbf{a} + \lambda\mathbf{u} + \mu\mathbf{v}$$

where $\mathbf{u}$, $\mathbf{v}$ are two non-parallel direction vectors in the plane; $\mathbf{n} = \mathbf{u} \times \mathbf{v}$.

6. Intersections and Angles

Line and Plane

Substitute the line's parametric equations into $\mathbf{r} \cdot \mathbf{n} = d$ and solve for λ .

- One solution $\Rightarrow$ a single intersection point.
- No solution $\Rightarrow$ line is parallel to the plane (and not in it).
- Identity $\Rightarrow$ line lies in the plane.

Angle between Line and Plane

$$\sin \theta = \frac{|\mathbf{d} \cdot \mathbf{n}|}{|\mathbf{d}||\mathbf{n}|}$$

(θ is the angle between the line and its projection onto the plane.)

Angle between Two Planes

$$\cos \theta = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{|\mathbf{n}_1||\mathbf{n}_2|}$$

Shortest Distance

Between	Formula
Point $\mathbf{p}$ and plane $\mathbf{r} \cdot \mathbf{n} = d$	$\frac{ \mathbf{p} \cdot \mathbf{n} - d }{ \mathbf{n} }$
Point $\mathbf{p}$ and line $\mathbf{r} = \mathbf{a} + \lambda \mathbf{d}$	$\frac{ (\mathbf{p} - \mathbf{a}) \times \mathbf{d} }{ \mathbf{d} }$
Two skew lines	$\frac{ (\mathbf{a}_2 - \mathbf{a}_1) \cdot (\mathbf{d}_1 \times \mathbf{d}_2) }{ \mathbf{d}_1 \times \mathbf{d}_2 }$

Distance from $P(4, 1, 2)$ to plane $2x - y + 2z = 5$.

$\mathbf{n} = (2, -1, 2)$, $|\mathbf{n}| = 3$. $\mathbf{p} \cdot \mathbf{n} - d = 8 - 1 + 4 - 5 = 6$. Distance = $6/3 = 2$.

7. Relationship Between Two Lines

Two lines in 3D may be:

- **Parallel** — direction vectors proportional.
- **Intersecting** — a single common point.
- **Skew** — neither parallel nor intersecting.
- **Coincident** — same line.

To test: solve the parametric equations component by component for λ, μ . If the x - and y -equations give consistent (λ, μ) that also satisfy the z -equation $\Rightarrow$ lines meet; otherwise skew.

8. Exam Style Strategy

- Always state the direction vector and a point when writing a line in vector form.
- For cross products, use the determinant layout — faster and fewer errors.
- Check whether three points are collinear by computing $(\mathbf{B} - \mathbf{A}) \times (\mathbf{C} - \mathbf{A})$ — zero iff collinear.
- Check coplanarity of four points by taking a scalar triple product — zero iff coplanar.